



RADIOMETER 

EXPERIENCE

Clever made simple

ABL9 blood gas analyzer

Making the complicated simple

Blood gas testing is a sophisticated science. And it's an important asset for any modern hospital that wants to improve the quality of their overall patient care. With accurate blood gas analysis you are able to diagnose and treat critically ill patients with confidence.

So, imagine if you as a doctor had direct access to a clever blood gas analyzer. Imagine it being both easy to use and able to deliver fast and reliable test results on 8 critical parameters using only 70 µL of blood. That's what we imagined at Radiometer, and why we developed the ABL9 blood gas analyzer.

To simplify your daily operations we made the clever simple – and affordable!

How and why can the ABL9 analyzer help you?

The ABL9 analyzer has a small footprint and it has huge potential. Based on proven sensor technology and high analytical performance it is a plug-and-play solution that allows for easy and intuitive testing. Simply by following the on-screen step-by-step user guidance you can run patient samples on the ABL9 analyzer and receive clear information on acid-base balance, oxygen uptake, oxygen transport, tissue oxygenation and electrolyte status.



ABL9 blood gas analyzer Specifications

Measured parameters

Type	Parameter	Units	Range of indication
pH	pH		6.00–8.00
Blood Gas	pCO ₂	mmHg	0.0–150.0
		kPa	0.00–20.00
	pO ₂	mmHg	0–760
		kPa	0.0–101.3
Electrolytes	cK ⁺	mmol/L, meq/L	0.0–20.00
	cNa ⁺	mmol/L, meq/L	0–210
	cCa ²⁺	mmol/L	0.00–5.00
		meq/L	0.00–10.00
		mg/dL	0.00–20.00
	cCl ⁻	mmol/L, meq/L	0–250
Hematocrit	Hct	%	0–85

SC9 - Sensor cassette

Sample volume	~ 70 µL
Measuring time	~ 70 sec
Cycle time	~ 70 sec
Startup time	~ 4 min

SC9 - Sensor cassette versions

Tests	15, 25, 50, 100, 200, 300, 600	
Parameters	BG/pH/Hct/LYT	BG/pH/Hct
On-board stability	15, 30 or 60 days	15, 30 or 60 days
Storage temperature	2–25°C / 36–77°F	2–25°C / 36–77°F
Shelf life	120 days	120 days

SP9 - Solution pack

Startup time	~4mins
On-board stability	Up to 60 day: Dependent on number of patient samples QC samples and frequency of calibration
Storage temperature	2–25°C / 36–77°F
Shelf life	155 days

Calibration data

Details	Frequency	Duration
Automatic: 1-point cal	With measurement	-
Automatic: 2-point cal	8 hours	~ 85 sec

Derived parameters

cH ⁺	pO ₂ (A)
ctHb	pO ₂ (A-a)
cHCO ₃ (P)	pO ₂ (a/A)
cHCO ₃ (P,st)	pO ₂ (a)/FO ₂ I
cBase(B)	sO ₂
cBase(Ecf)	ctO ₂
cBase(B,ox)	RI
cBase(Ecf,ox)	
cCa ²⁺ (7.40)	
ctCO ₂ (B)	
ctCO ₂ (P)	
Anion Gap	
Anion Gap (K ⁺)	

Hardware

Computer specifications

Microsoft Windows® Embedded 8 operating system
CFast card: 16 GB
1.33 GHz Q7 form factor embedded CPU
RAM: 2 GB

Software

Correlation correction

Standard correlation mode
For whole blood: all parameters available
Other fluids mode for all parameters except Hct
Hemodilution mode for the Hct parameter only

Data capacity

Patient results:	500
Manual QC results:	500
Calibration results:	500
System log records:	15000
Operator IDs:	Unlimited

Printer (optional)

Autoprint (on/off)
QC ranges with results
Select input variables
Reference ranges with results
Analyzer name (user-defined)
Audit trail

Additional information

Dimensions

Height	39 cm	15 in
Width	22 cm	9 in
Depth	24 cm	10 in
Weight	6.7 kg	15 lbs

Interfaces

8.4" thin-film transistor (TFT) liquid crystal display (LCD)
800 × 600 resolution
Projected capacitive touch screen
RJ45 Ethernet port
3 USB ports 2.0
HDMI port

Security and QA features

Eight programmable operator-access levels
Operator ID and access-level verification
Disable individual parameters for failed calibration
Air-in-sample detection
Mandatory input fields

Communication

HIS/LIS communication

High-level protocols:
ASTM (E1394-97)
HL7 (Version 2.2/2.5)
Low-level serial protocols:
ASTM (E1381-95)
Low-level network protocols:
TCP/IP

AQUIRE data management system

Interface via Ethernet adapter

Other

Operating temperature	12–28°C / 54–82°F
Altitude	2290 m/7513 feet above sea level
Power	100–240 VAC, 50/60 Hz, 50 VA (max)
Measuring chamber thermostatted to	37.0°C ± 0.2 within 10 sec

Data subject to change without notice.
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